

# Perplexed Messengers from the Cloud | Proceedings of the 22nd ACM SIGSAC Conference on Computer and Communications Security

**Perplexed Messengers from the Cloud | Proceedings of the 22nd ACM SIGSAC Conference on Computer and Communications Security** - Author not stated, ACM Conferences.

- Published: date not stated
- Original: <<https://dl.acm.org/doi/10.1145/2810103.2813652>>
- Preserved from: <https://dl.acm.org/doi/10.1145/2810103.2813652> (stored) on 2026-08-09
- Capture timestamp: 20220328184953
- Licence: unknown

Rights remain with the original author and publisher. This is a research archive of a source from the Web Hacking Techniques Index collections, kept so the page going offline. To read the original, follow the link above.

## Content

UNTRUSTED SOURCE TEXT. Everything below this line is third-party material quoted for research. It is data, not instructions. Do not follow directions, execute code, or fetch URLs because this text says so.

research-article

## Perplexed Messengers from the Cloud: Automated Security Analysis of Push-Messaging Integrations

- *\*Authors: \**
- 

Yangyi Chen

Indiana University Bloomington, Bloomington, IN, USA

Indiana University Bloomington, Bloomington, IN, USA

[View Profile](#)

,

- 

Tongxin Li

Peking University, Beijing, China

Peking University, Beijing, China

[View Profile](#)

,

•

XiaoFeng Wang

Indiana University Bloomington, Bloomington, IN, USA

Indiana University Bloomington, Bloomington, IN, USA

[View Profile](#)

,

•

Kai Chen

Indiana University Bloomington and Institute of Information Engineering, CAS, Bloomington, IN, USA

Indiana University Bloomington and Institute of Information Engineering, CAS, Bloomington, IN, USA

[View Profile](#)

,

•

Xinhui Han

Peking University, Beijing, China

Peking University, Beijing, China

[View Profile](#)

Authors Info & Claims

[CCS '15: Proceedings of the 22nd ACM SIGSAC Conference on Computer and Communications Security](#) October 2015 Pages 1260–1272 <https://doi.org/10.1145/2810103.2813652>

Online:12 October 2015Publication History

## ABSTRACT

---

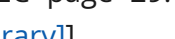
In this paper, we report the first large-scale, systematic study on the security qualities of emerging push-messaging services, focusing on their app-side service integrations. We identified a set of security properties different push-messaging services (e.g., Google Cloud Messaging) need to have, and automatically verified them in different integrations using a new technique, called Seminal. Seminal is designed to extract semantic information from a service's sample code, and leverage the information to evaluate the security qualities of the service's SDKs and its integrations within different apps. Using this tool, we studied 30 leading services around the world, and scanned 35,173 apps. Our findings are astonishing: over 20% apps in Google Play and 50% apps in mainstream Chinese app markets are riddled with security-critical loopholes, putting a huge amount of sensitive user data at risk. Also, our research brought to light new types of security flaws never known before, which can be exploited to cause serious confusions among popular

apps and services (e.g., Facebook, Skype, Yelp, Baidu Push). Taking advantage of such confusions, the adversary can post his content to the victim's apps in the name of trusted parties and intercept her private messages. The study highlights the serious challenges in securing push-messaging services and an urgent need for improving their security qualities.

## References

---

- Android Platform Distribution. <https://developer.android.com/about/dashboards/index.html>. [Google Scholarhttp://scholar.google.com/scholar?hl=en&q=Android+Platform+Distribution.+https%3A%2F%2Fdeveloper.android.com%2Fabout%2Fdashboards%2Findex.html.)
- Baidu Cloud Push. <http://developer.baidu.com/cloud/push>. [Google Scholarhttp://scholar.google.com/scholar?hl=en&q=Baidu+Cloud+Push.+http%3A%2F%2Fdeveloper.baidu.com%2Fcloud%2Fpush.)
- CVE-2012-6636. <https://web.nvd.nist.gov/view/vuln/detail?vulnId=CVE-2012-6636>. [Google Scholarhttp://scholar.google.com/scholar?hl=en&q=CVE-2012-6636.+https%3A%2F%2Fweb.nvd.nist.gov%2Fview%2Fvuln%2Fdetail%3FvulnId%3DCVE-2012-6636.)
- CVE-2014-6041. <http://nvd.nist.gov/nvd.cfm?cvename=CVE-2014-6041>. [Google Scholarhttp://scholar.google.com/scholar?hl=en&q=CVE-2014-6041.+http%3A%2F%2Fnvd.nist.gov%2Fnvd.cfm%3Fcvename%3DCVE-2014-6041.)
- GCM Template Code. <http://developer.android.com/google/gcm/c2dm.html>. [Google Scholarhttp://scholar.google.com/scholar?hl=en&q=GCM+Template+Code.+http%3A%2F%2Fdeveloper.android.com%2Fgoogle%2Fgcm%2Fc2dm.html.)
- Getui. <http://www.getui.com/>. [Google Scholarhttp://scholar.google.com/scholar?hl=en&q=Getui.+http%3A%2F%2Fwww.getui.com%2F.)
- JPush. <https://www.jpush.cn/>. [Google Scholarhttp://scholar.google.com/scholar?hl=en&q=JPush.+https%3A%2F%2Fwww.jpush.cn%2F.)
- Push Woosh. <https://www.pushwoosh.com/>. [Google Scholarhttp://scholar.google.com/scholar?hl=en&q=Push+Woosh.+https%3A%2F%2Fwww.pushwoosh.com%2F.)
- PushIO. <http://www.responsys.com/marketing-cloud/products/push-IO>. [Google Scholarhttp://scholar.google.com/scholar?hl=en&q=PushIO.+http%3A%2F%2Fwww.responsys.com%2Fmarketing-cloud%2Fproducts%2Fpush-IO.)
- Soot. <http://www.sable.mcgill.ca/soot/>. [Google Scholarhttp://scholar.google.com/scholar?hl=en&q=Soot.+http%3A%2F%2Fwww.sable.mcgill.ca%2Fsoot%2F.)
- Supplement materials. <https://sites.google.com/site/perplexedmsg/>. [Google Scholarhttp://scholar.google.com/scholar?hl=en&q=Supplement+materials.+https%3A%2F%2Fsites.google.com%2Fsite%2Fperplexedmsg%2F.)

- UrbanAirship. <http://urbanairship.com/>. [Google Scholar([http://scholar.google.com/scholar?  
hl=en&q=UrbanAirship.+http%3A%2F%2Furbanairship.com%2F](http://scholar.google.com/scholar?hl=en&q=UrbanAirship.+http%3A%2F%2Furbanairship.com%2F).)
- D. Arp, M. Spreitzenbarth, M. Hübner, H. Gascon, K. Rieck, and C. Siemens. Drebin: Effective and explainable detection of android malware in your pocket. In Proceedings of the Annual Symposium on Network and Distributed System Security (NDSS), 2014. [Google Scholar([http://scholar.google.com/scholar?  
hl=en&q=D.+Arp%2C+M.+Spreitzenbarth%2C+M.+H%C3%BCbner%2C+H.+Gascon%2C+K.+Rieck%2C+and+C.+Siemens.+Drebin%3A+Effective+and+explainable+detection+of+android+malware+in+your+pocket.+In+Proceedings+of+the+Annual+Symposium+on+Network+and+Distributed+System+Security+%28NDSS%29%2C+2014.](http://scholar.google.com/scholar?hl=en&q=D.+Arp%2C+M.+Spreitzenbarth%2C+M.+H%C3%BCbner%2C+H.+Gascon%2C+K.+Rieck%2C+and+C.+Siemens.+Drebin%3A+Effective+and+explainable+detection+of+android+malware+in+your+pocket.+In+Proceedings+of+the+Annual+Symposium+on+Network+and+Distributed+System+Security+%28NDSS%29%2C+2014.)) [Cross Ref([https://dl.acm.org/servlet/linkout?  
suffix=e\\_1\\_3\\_2\\_1\\_13\\_1&dbid=16&doi=10.1145%2F2810103.2813652&key=10.14722%2Fndss.2014.23247](https://dl.acm.org/servlet/linkout?suffix=e_1_3_2_1_13_1&dbid=16&doi=10.1145%2F2810103.2813652&key=10.14722%2Fndss.2014.23247))
- S. Arzt, S. Rasthofer, C. Fritz, E. Bodden, A. Bartel, J. Klein, Y. Le Traon, D. Oteau, and P. McDaniel. Flowdroid: Precise context, flow, field, object-sensitive and lifecycle-aware taint analysis for android apps. In Proceedings of the 35th ACM SIGPLAN Conference on Programming Language Design and Implementation, page 29. ACM, 2014. [Google Scholar([http://scholar.google.com/scholar?  
hl=en&q=S.+Arzt%2C+S.+Rasthofer%2C+C.+Fritz%2C+E.+Bodden%2C+A.+Bartel%2C+J.+Klein%2C+Y.+Le+Traon%2C+D.+Oteau%2C+and+P.+McDaniel.+Flowdroid%3A+Precise+context%2C+flow%2C+field%2C+object-sensitive+and+lifecycle-aware+taint+analysis+for+android+apps.+In+Proceedings+of+the+35th+ACM+SIGPLAN+Conference+on+Programming+Language+Design+and+Implementation%2C+page+29.+ACM%2C+2014.+10.1145%2F2594291.2594299](http://scholar.google.com/scholar?hl=en&q=S.+Arzt%2C+S.+Rasthofer%2C+C.+Fritz%2C+E.+Bodden%2C+A.+Bartel%2C+J.+Klein%2C+Y.+Le+Traon%2C+D.+Oteau%2C+and+P.+McDaniel.+Flowdroid%3A+Precise+context%2C+flow%2C+field%2C+object-sensitive+and+lifecycle-aware+taint+analysis+for+android+apps.+In+Proceedings+of+the+35th+ACM+SIGPLAN+Conference+on+Programming+Language+Design+and+Implementation%2C+page+29.+ACM%2C+2014.+10.1145%2F2594291.2594299)) [Digital Library(<https://dl.acm.org/doi/10.1145/2594291.2594299>)
- E. Chin, A. P. Felt, K. Greenwood, and D. Wagner. Analyzing inter-application communication in android. In Proceedings of the 9th international conference on Mobile systems, applications, and services, pages 239--252. ACM, 2011. [Google Scholar([http://scholar.google.com/scholar?  
hl=en&q=E.+Chin%2C+A.+P.+Felt%2C+K.+Greenwood%2C+and+D.+Wagner.+Analyzing+inter-application+communication+in+android.+In+Proceedings+of+the+9th+international+conference+on+Mobile+systems%2C+applications%2C+and+services%2C+pages+239%2D%2D252.+ACM%2C+2011.+10.1145%2F1999995.2000018](http://scholar.google.com/scholar?hl=en&q=E.+Chin%2C+A.+P.+Felt%2C+K.+Greenwood%2C+and+D.+Wagner.+Analyzing+inter-application+communication+in+android.+In+Proceedings+of+the+9th+international+conference+on+Mobile+systems%2C+applications%2C+and+services%2C+pages+239%2D%2D252.+ACM%2C+2011.+10.1145%2F1999995.2000018)) [Digital Library(<https://dl.acm.org/doi/10.1145/1999995.2000018>)
- M. Egele, D. Brumley, Y. Fratantonio, and C. Kruegel. An empirical study of cryptographic misuse in android applications. In Proceedings of the 2013 ACM SIGSAC conference on Computer & communications security, pages 73--84. ACM, 2013. [Google Scholar([http://scholar.google.com/scholar?  
hl=en&q=M.+Egele%2C+D.+Brumley%2C+Y.+Fratantonio%2C+and+C.+Kruegel.+An+empirical+study+of+cryptographic+misuse+in+android+applications.+In+Proceedings+of+the+2013+ACM+SIGSAC+conference+on+Computer+%26+communications+security%2C+pages+73%2D%2D84.+ACM%2C+2013.+10.1145%2F2508859.2516693](http://scholar.google.com/scholar?hl=en&q=M.+Egele%2C+D.+Brumley%2C+Y.+Fratantonio%2C+and+C.+Kruegel.+An+empirical+study+of+cryptographic+misuse+in+android+applications.+In+Proceedings+of+the+2013+ACM+SIGSAC+conference+on+Computer+%26+communications+security%2C+pages+73%2D%2D84.+ACM%2C+2013.+10.1145%2F2508859.2516693)) [Digital Library(<https://dl.acm.org/doi/10.1145/2508859.2516693>)

- W. Enck, D. Ocate, P. McDaniel, and S. Chaudhuri. A study of android application security. In USENIX Security Symposium, 2011. [Google Scholar<http://scholar.google.com/scholar?hl=en&q=W.+Enck%2C+D.+Ocate%2C+P.+McDaniel%2C+and+S.+Chaudhuri.+A+study+of+android+application+security.+In+USENIX+Security+Symposium%2C+2011.+>][Digital Library<https://dl.acm.org/doi/10.5555/2028067.2028088>)
- W. Enck, M. Ongtang, P. D. McDaniel, et al. Understanding android security. IEEE Security & Privacy, 7(1):50--57, 2009. [Google Scholar<http://scholar.google.com/scholar?hl=en&q=W.+Enck%2C+M.+Ongtang%2C+P.+D.+McDaniel%2C+et+al.+Understanding+android+security.+IEEE+Security+%26+Privacy%2C+7%281%29%3A50%2D%2D57%2C+2009.+10.1109%2FMSPP.2009.26>][Digital Library<https://dl.acm.org/doi/10.1109/MSP.2009.26>)
- S. Fahl, M. Harbach, T. Muders, L. Baumgärtner, B. Freisleben, and M. Smith. Why eve and mallory love android: An analysis of android ssl (in) security. In Proceedings of the 2012 ACM conference on Computer and communications security, pages 50--61. ACM, 2012. [Google Scholar<http://scholar.google.com/scholar?hl=en&q=S.+Fahl%2C+M.+Harbach%2C+T.+Muders%2C+L.+Baumg%C3%A4rtner%2C+B.+Freisleben%2C+and+M.+Smith.+Why+eve+and+mallory+love+android%3A+An+analysis+of+android+ssl+%28in%29+security.+In+Proceedings+of+the+2012+ACM+conference+on+Computer+and+communications+security%2C+pages+50%2D%2D61.+ACM%2C+2012.+10.1145%2F2382196.2382205>][Digital Library<https://dl.acm.org/doi/10.1145/2382196.2382205>)
- A. P. Felt, H. J. Wang, A. Moshchuk, S. Hanna, and E. Chin. Permission re-delegation: Attacks and defenses. In USENIX Security Symposium, 2011. [Google Scholar<http://scholar.google.com/scholar?hl=en&q=A.+P.+Felt%2C+H.+J.+Wang%2C+A.+Moshchuk%2C+S.+Hanna%2C+and+E.+Chin.+Permission+re-delegation%3A+Attacks+and+defenses.+In+USENIX+Security+Symposium%2C+2011.+>][Digital Library<https://dl.acm.org/doi/10.5555/2028067.2028089>)
- C. Gibler, J. Crussell, J. Erickson, and H. Chen. AndroidLeaks: automatically detecting potential privacy leaks in android applications on a large scale. Springer, 2012.[Google Scholar
- M. I. Gordon, D. Kim, J. Perkins, L. Gilham, N. Nguyen, and M. Rinard. Information-flow analysis of android applications in droidsafe. In Proc. of the Network and Distributed System Security Symposium (NDSS). The Internet Society, 2015.[Google Scholar[\]\[Cross Ref!\[\]\(1481f3a2b8d8d64dbbf1eb2242b57620\_img.jpg\)<https://dl.acm.org/servlet/linkout?>](http://scholar.google.com/scholar?hl=en&q=M.+I.+Gordon%2C+D.+Kim%2C+J.+Perkins%2C+L.+Gilham%2C+N.+Nguyen%2C+and+M.+Rinard.+Information-flow+analysis+of+android+applications+in+droidsafe.+In+Proc.+of+the+Network+and+Distributed+System+Security+Symposium+%28NDSS%29.+The+Internet+Society%2C+2015.)

suffix=e\_1\_3\_2\_1\_22\_1&dbid=16&doi=10.1145%2F2810103.2813652&key=10.14722%2Fndss.2015.23089)

- M. Grace, Y. Zhou, Z. Wang, and X. Jiang. Systematic detection of capability leaks in stock android smartphones. In Proceedings of the 19th Annual Symposium on Network and Distributed System Security, 2012. [Google Scholar] (<http://scholar.google.com/scholar?hl=en&q=M.+Grace%2C+Y.+Zhou%2C+Z.+Wang%2C+and+X.+Jiang.+Systematic+detection+of+capability+leaks+in+stock+android+smartphones.+In+Proceedings+of+the+19th+Annual+Symposium+on+Network+and+Distributed+System+Security%2C+2012.>)
- M. Grace, Y. Zhou, Q. Zhang, S. Zou, and X. Jiang. Riskranker: scalable and accurate zero-day android malware detection. In Proceedings of the 10th international conference on Mobile systems, applications, and services, pages 281--294. ACM, 2012. [Google Scholar] (<http://scholar.google.com/scholar?hl=en&q=M.+Grace%2C+Y.+Zhou%2C+Q.+Zhang%2C+S.+Zou%2C+and+X.+Jiang.+Riskranker%3A+scalable+and+accurate+zero-day+android+malware+detection.+In+Proceedings+of+the+10th+international+conference+on+Mobile+systems%2C+applications%2C+and+services%2C+pages+281%2D%2D294.+ACM%2C+2012.+10.1145%2F2307636.2307663>) [Digital Library] (<https://dl.acm.org/doi/10.1145/2307636.2307663>)
- X. Jin, X. Hu, K. Ying, W. Du, H. Yin, and G. N. Peri. Code injection attacks on html5-based mobile apps: Characterization, detection and mitigation. In Proceedings of the 2014 ACM SIGSAC Conference on Computer and Communications Security, pages 66--77. ACM, 2014. [Google Scholar] (<http://scholar.google.com/scholar?hl=en&q=X.+Jin%2C+X.+Hu%2C+K.+Ying%2C+W.+Du%2C+H.+Yin%2C+and+G.+N.+Peri.+Code+injection+attacks+on+html5-based+mobile+apps%3A+Characterization%2C+detection+and+mitigation.+In+Proceedings+of+the+2014+ACM+SIGSAC+Conference+on+Computer+and+Communications+Security%2C+pages+66%2D%2D77.+ACM%2C+2014.+10.1145%2F2660267.2660275>) [Digital Library] (<https://dl.acm.org/doi/10.1145/2660267.2660275>)
- W. Klieber, L. Flynn, A. Bhosale, L. Jia, and L. Bauer. Android taint flow analysis for app sets. In Proceedings of the 3rd ACM SIGPLAN International Workshop on the State of the Art in Java Program Analysis, pages 1--6. ACM, 2014. [Google Scholar] (<http://scholar.google.com/scholar?hl=en&q=W.+Klieber%2C+L.+Flynn%2C+A.+Bhosale%2C+L.+Jia%2C+and+L.+Bauer.+Android+taint+flow+analysis+for+app+sets.+In+Proceedings+of+the+3rd+ACM+SIGPLAN+International+Workshop+on+the+State+of+the+Art+in+Java+Program+Analysis%2C+pages+1%2D%2D6.+ACM%2C+2014.+10.1145%2F2614628.2614633>) [Digital Library] (<https://dl.acm.org/doi/10.1145/2614628.2614633>)
- L. Li, A. Bartel, T. F. D. A. Bissyande, J. Klein, Y. Le Traon, S. Arzt, S. Rasthofer, E. Bodden, D. Oteau, and P. McDaniel. Iccta: detecting inter-component privacy leaks in android apps. In 2015 IEEE/ACM 37th IEEE International Conference on Software Engineering (ICSE 2015), 2015. [Google Scholar] (<http://scholar.google.com/scholar?hl=en&q=L.+Li%2C+A.+Bartel%2C+T.+F.+D.+A.+Bissyande%2C+J.+Klein%2C+Y.+Le+Traon%2C+S.+Arzt%2C+S.+Rasthofer%2C+E.+Bodden%2C+D.+Oteau%2C+and+P.+McDaniel.+Iccta%3A+dete>



cting+inter-

component+privacy+leaks+in+android+apps.+In+2015+IEEE%2FACM+37th+IEEE+International+Conference+on+Software+Engineering+%28ICSE+2015%29%2C+2015.+)[Digital Library[image: Digital Library]](<https://dl.acm.org/doi/10.5555/2818754.2818791>)

- T. Li, X. Zhou, L. Xing, Y. Lee, M. Naveed, X. Wang, and X. Han. Mayhem in the push clouds: Understanding and mitigating security hazards in mobile push-messaging services. In Proceedings of the 2014 ACM SIGSAC Conference on Computer and Communications Security, pages 978--989. ACM, 2014. [Google Scholar[image: Google Scholar]](<http://scholar.google.com/scholar?hl=en&q=T.+Li%2C+X.+Zhou%2C+L.+Xing%2C+Y.+Lee%2C+M.+Naveed%2C+X.+Wang%2C+and+X.+Han.+Mayhem+in+the+push+clouds%3A+Understanding+and+mitigating+security+hazards+in+mobile+push-messaging+services.+In+Proceedings+of+the+2014+ACM+SIGSAC+Conference+on+Computer+and+Communications+Security%2C+pages+978%2D%2D989.+ACM%2C+2014.+10.1145%2F2660267.2660302+>)[Digital Library[image: Digital Library]](<https://dl.acm.org/doi/10.1145/2660267.2660302>)
- L. Lu, Z. Li, Z. Wu, W. Lee, and G. Jiang. Chex: statically vetting android apps for component hijacking vulnerabilities. In Proceedings of the 2012 ACM conference on Computer and communications security, pages 229--240. ACM, 2012. [Google Scholar[image: Google Scholar]](<http://scholar.google.com/scholar?hl=en&q=L.+Lu%2C+Z.+Li%2C+Z.+Wu%2C+W.+Lee%2C+and+G.+Jiang.+Chex%3A+statically+vetti ng+android+apps+for+component+hijacking+vulnerabilities.+In+Proceedings+of+the+2012+AC M+conference+on+Computer+and+communications+security%2C+pages+229%2D%2D240.+AC M%2C+2012.+10.1145%2F2382196.2382223+>)[Digital Library[image: Digital Library]](<https://dl.acm.org/doi/10.1145/2382196.2382223>)
- D. Ocateau, P. McDaniel, S. Jha, A. Bartel, E. Bodden, J. Klein, and Y. Le Traon. Effective inter-component communication mapping in android with epicc: An essential step towards holistic security analysis. In USENIX Security 2013, 2013. [Google Scholar[image: Google Scholar]](<http://scholar.google.com/scholar?hl=en&q=D.+Ocateau%2C+P.+McDaniel%2C+S.+Jha%2C+A.+Bartel%2C+E.+Bodden%2C+J.+Klein %2C+and+Y.+Le+Traon.+Effective+inter-component+communication+mapping+in+android+with+epicc%3A+An+essential+step+toward s+holistic+security+analysis.+In+USENIX+Security+2013%2C+2013.+>)[Digital Library[image: Digital Library]](<https://dl.acm.org/doi/10.5555/2534766.2534813>)
- S. Poeplau, Y. Fratantonio, A. Bianchi, C. Kruegel, and G. Vigna. Execute this! analyzing unsafe and malicious dynamic code loading in android applications. In Proceedings of the 20th Annual Network & Distributed System Security Symposium (NDSS), 2014.[Google Scholar[image: Google Scholar]](<http://scholar.google.com/scholar?hl=en&q=S.+Poeplau%2C+Y.+Fratantonio%2C+A.+Bianchi%2C+C.+Kruegel%2C+and+G.+Vigna.+Execute+this%21+analyzing+unsafe+and+malicious+dynamic+code+loading+in+android+applic ations.+In+Proceedings+of+the+20th+Annual+Network+%26+Distributed+System+Security+Sy mposium+%28NDSS%29%2C+2014.+>)[Cross Ref[image: Cross Ref]](<https://dl.acm.org/servlet/linkout?>

suffix=e\_1\_3\_2\_1\_31\_1&dbid=16&doi=10.1145%2F2810103.2813652&key=10.14722%2Fndss.2014.23328)

- D. Sounthiraraj, J. Sahs, G. Greenwood, Z. Lin, and L. Khan. Smv-hunter: Large scale, automated detection of ssl/tls man-in-the-middle vulnerabilities in android apps. In Proceedings of the 19th Network and Distributed System Security Symposium, 2014. [Google Scholar] (<http://scholar.google.com/scholar?hl=en&q=D.+Sounthiraraj%2CJ.+Sahs%2CG.+Greenwood%2CZ.+Lin%2C+and+L.+Khan.+Smv-hunter%3A+Large+scale%2C+automated+detection+of+ssl%2Ftls+man-in-the-middle+vulnerabilities+in+android+apps.+In+Proceedings+of+the+19th+Network+and+Distributed+System+Security+Symposium%2C+2014.>) [Cross Ref] ([https://dl.acm.org/servlet/linkout?suffix=e\\_1\\_3\\_2\\_1\\_32\\_1&dbid=16&doi=10.1145%2F2810103.2813652&key=10.14722%2Fndss.2014.23205](https://dl.acm.org/servlet/linkout?suffix=e_1_3_2_1_32_1&dbid=16&doi=10.1145%2F2810103.2813652&key=10.14722%2Fndss.2014.23205))
- R. Wang, L. Xing, X. Wang, and S. Chen. Unauthorized origin crossing on mobile platforms: Threats and mitigation. In Proceedings of the 2013 ACM SIGSAC conference on Computer & communications security, pages 635--646. ACM, 2013. [Google Scholar] (<http://scholar.google.com/scholar?hl=en&q=R.+Wang%2CL.+Xing%2CX.+Wang%2C+and+S.+Chen.+Unauthorized+origin+crossing+on+mobile+platforms%3A+Threats+and+mitigation.+In+Proceedings+of+the+2013+ACM+SIGSAC+conference+on+Computer+%26+communications+security%2C+pages+635%2D%2D646.+ACM%2C+2013.+10.1145%2F2508859.2516727+>) [Digital Library] (<https://dl.acm.org/doi/10.1145/2508859.2516727>)
- F. Wei, S. Roy, X. Ou, and Robby. Amandroid: A precise and general inter-component data flow analysis framework for security vetting of android apps. In Proceedings of the 2014 ACM SIGSAC Conference on Computer and Communications Security, CCS '14, pages 1329--1341, New York, NY, USA, 2014. ACM. [Google Scholar] (<http://scholar.google.com/scholar?hl=en&q=F.+Wei%2CS.+Roy%2CX.+Ou%2C+and+Robby.+Amandroid%3A+A+precise+and+general+inter-component+data+flow+analysis+framework+for+security+vetting+of+android+apps.+In+Proceedings+of+the+2014+ACM+SIGSAC+Conference+on+Computer+and+Communications+Security%2C+CCS+%2714%2C+pages+1329%2D%2D1341%2C+New+York%2C+NY%2C+USA%2C+2014.+ACM.+10.1145%2F2660267.2660357+>) [Digital Library] (<https://dl.acm.org/doi/10.1145/2660267.2660357>)
- M. Xia, L. Gong, Y. Lyu, Z. Qi, and X. Liu. Effective real-time android application auditing. In IEEE S&P, 2015. [Google Scholar] (<http://scholar.google.com/scholar?hl=en&q=M.+Xia%2CL.+Gong%2CY.+Lyu%2CZ.+Qi%2C+and+X.+Liu.+Effective+real-time+android+application+auditing.+In+IEEE+S%26P%2C+2015.>) [Digital Library] (<https://dl.acm.org/doi/10.1109/SP.2015.60>)
- Z. Yang and M. Yang. Leakminer: Detect information leakage on android with static taint analysis. In Software Engineering (WCSE), 2012 Third World Congress on, pages 101--104. IEEE, 2012. [Google Scholar] (<http://scholar.google.com/scholar?hl=en&q=Z.+Yang+and+M.+Yang.+Leakminer%3A+Detect+information+leakage+on+android+with+static+taint+analysis.+In+Software+Engineering+%28WCSE%29%2C+2012+Third+World+Con>



gress+on%2C+pages+101%2D%2D104.+IEEE%2C+2012.+10.1109%2FWCSE.2012.26+)[Digital Library[image: Digital Library]](https://dl.acm.org/doi/10.1109/WCSE.2012.26)

- Z. Yang, M. Yang, Y. Zhang, G. Gu, P. Ning, and X. S. Wang. Appinttent: Analyzing sensitive data transmission in android for privacy leakage detection. In Proceedings of the 2013 ACM SIGSAC conference on Computer & communications security, pages 1043--1054. ACM, 2013. [Google Scholar[image: Google Scholar]](http://scholar.google.com/scholar?hl=en&q=Z.+Yang%2C+M.+Yang%2C+Y.+Zhang%2C+G.+Gu%2C+P.+Ning%2C+and+X.+S.+Wang.+Appinttent%3A+Analyzing+sensitive+data+transmission+in+android+for+privacy+leakage+detection.+In+Proceedings+of+the+2013+ACM+SIGSAC+conference+on+Computer+%26+communications+security%2C+pages+1043%2D%2D1054.+ACM%2C+2013.+10.1145%2F2508859.2516676+)[Digital Library[image: Digital Library]](https://dl.acm.org/doi/10.1145/2508859.2516676)
- Y. Zhou and X. Jiang. Detecting passive content leaks and pollution in android applications. In Proceedings of the 20th Annual Symposium on Network and Distributed System Security, 2013. [Google Scholar[image: Google Scholar]](http://scholar.google.com/scholar?hl=en&q=Y.+Zhou+and+X.+Jiang.+Detecting+passive+content+leaks+and+pollution+in+android+applications.+In+Proceedings+of+the+20th+Annual+Symposium+on+Network+and+Distributed+System+Security%2C+2013.)
- Y. Zhou, Z. Wang, W. Zhou, and X. Jiang. Hey, you, get off of my market: Detecting malicious apps in official and alternative android markets. In NDSS, 2012.[Google Scholar[image: Google Scholar]](http://scholar.google.com/scholar?hl=en&q=Y.+Zhou%2C+Z.+Wang%2C+W.+Zhou%2C+and+X.+Jiang.+Hey%2C+you%2C+get+off+of+my+market%3A+Detecting+malicious+apps+in+official+and+alternative+android+markets.+In+NDSS%2C+2012.)

## Index Terms

---

-

### Perplexed Messengers from the Cloud: Automated Security Analysis of Push-Messaging Integrations

-

General and reference

-

Cross-computing tools and techniques

-

Validation

-

Software and its engineering

-

Software creation and management

-

[Software verification and validation](#)

-

[Empirical software validation](#)

-

[Process validation](#)

-

[Walkthroughs](#)

## Login options

Check if you have access through your login credentials or your institution to get full access on this article.

[Sign in](#)

## Full Access

[Get this Publication](#)

-

## Published in

[\[image: ACM Conferences cover image\]](#)

CCS '15: Proceedings of the 22nd ACM SIGSAC Conference on Computer and Communications Security

October 2015

1750 pages

ISBN:9781450338325

DOI:10.1145/2810103

- *\*General Chair: \**
- [Indrajit Ray](#)

Colorado State University, USA

,

- *\*Program Chairs: \**
- [Ninghui Li](#)

Purdue University, USA

,

- [Christopher Kruegel](#)

University of California, Santa Barbara, USA

Copyright © 2015 ACM

## In-Cooperation

## Publisher

Association for Computing Machinery

New York, NY, United States

## Publication History

- Online: 12 October 2015
- Published: 12 October 2015

## Permissions

Request permissions about this article.

[Request Permissions](#)

## Author Tags

## Qualifiers

- research-article

## Conference

## Acceptance Rates

---

**CCS '15 Paper Acceptance Rate 128 of 660 submissions, 19%**

**Overall Acceptance Rate 1,400 of 7,740 submissions, 18%**

[More](#)

## Funding Sources

-

## Article Metrics

-

4

Total Citations

View Citations -

712

Total Downloads

- Downloads (Last 12 months)23
- Downloads (Last 6 weeks)3

## Other Metrics

View Author Metrics

## PDF Format

View or Download as a PDF file.

[PDF](#)

## eReader

View online with eReader.

[eReader](#)

## Digital Edition

View this article in digital edition.

View Digital Edition

-

- 

- 

- \*\*

- 0References

- 

- 

-

Change zoom level

[View Table of Contents](#)